

UNIVERSITÀ DEGLI STUDI
DI MILANO

SINGLE-CELL RNA SEQUENCING ANALYSIS OF Mus musculus VENTRAL MIDBRAIN CELLS

Pogliana Giacomo

Dataset overview

Molecular Architecture of the Mouse Nervous System

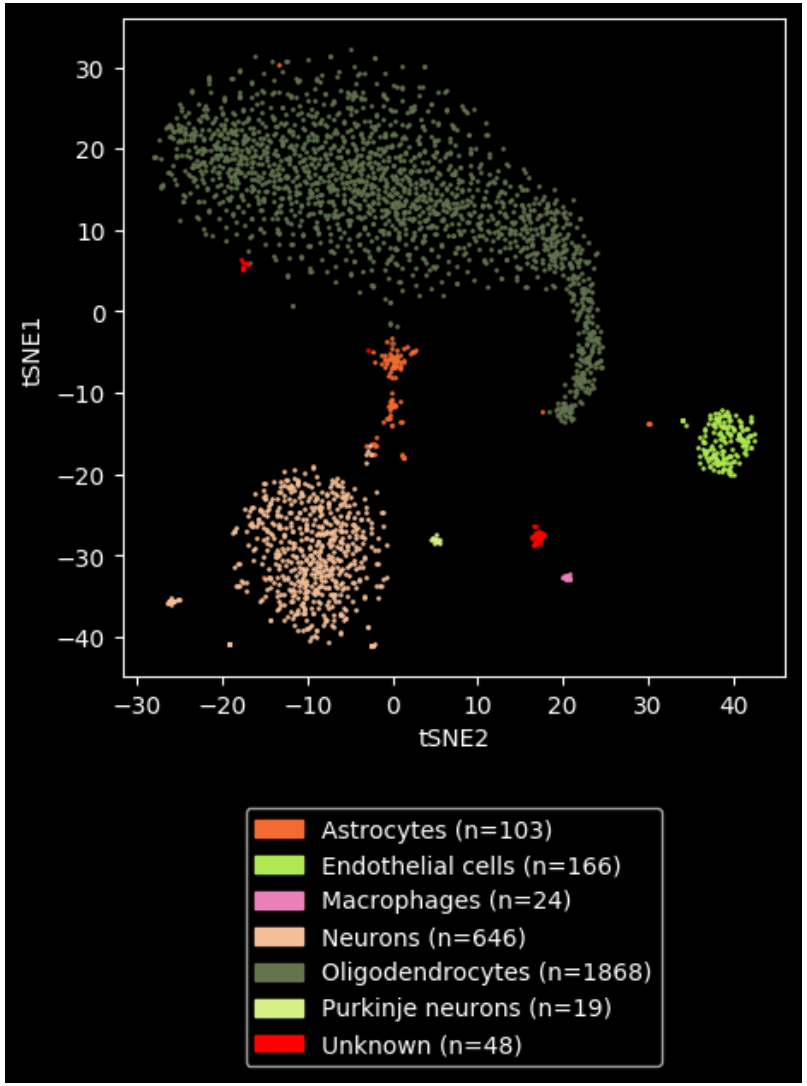
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Affiliations + expand

PMID: 30096314 PMCID: PMC6086934 DOI: 10.1016/j.cell.2018.06.021

PanglaoDB

SRA: SRA667466
SRS: SRS3059969
SRR: SRR6854153
Species: Mus musculus
Sample origin: Adolescent Mus musculus (P21, femmina, ceppo CD-1)
Protocol: 10x chromium
Instrument: Illumina HiSeq 2500
Number of cells: 3,028
Number of clusters: 12
Tissue: Ventral midbrain



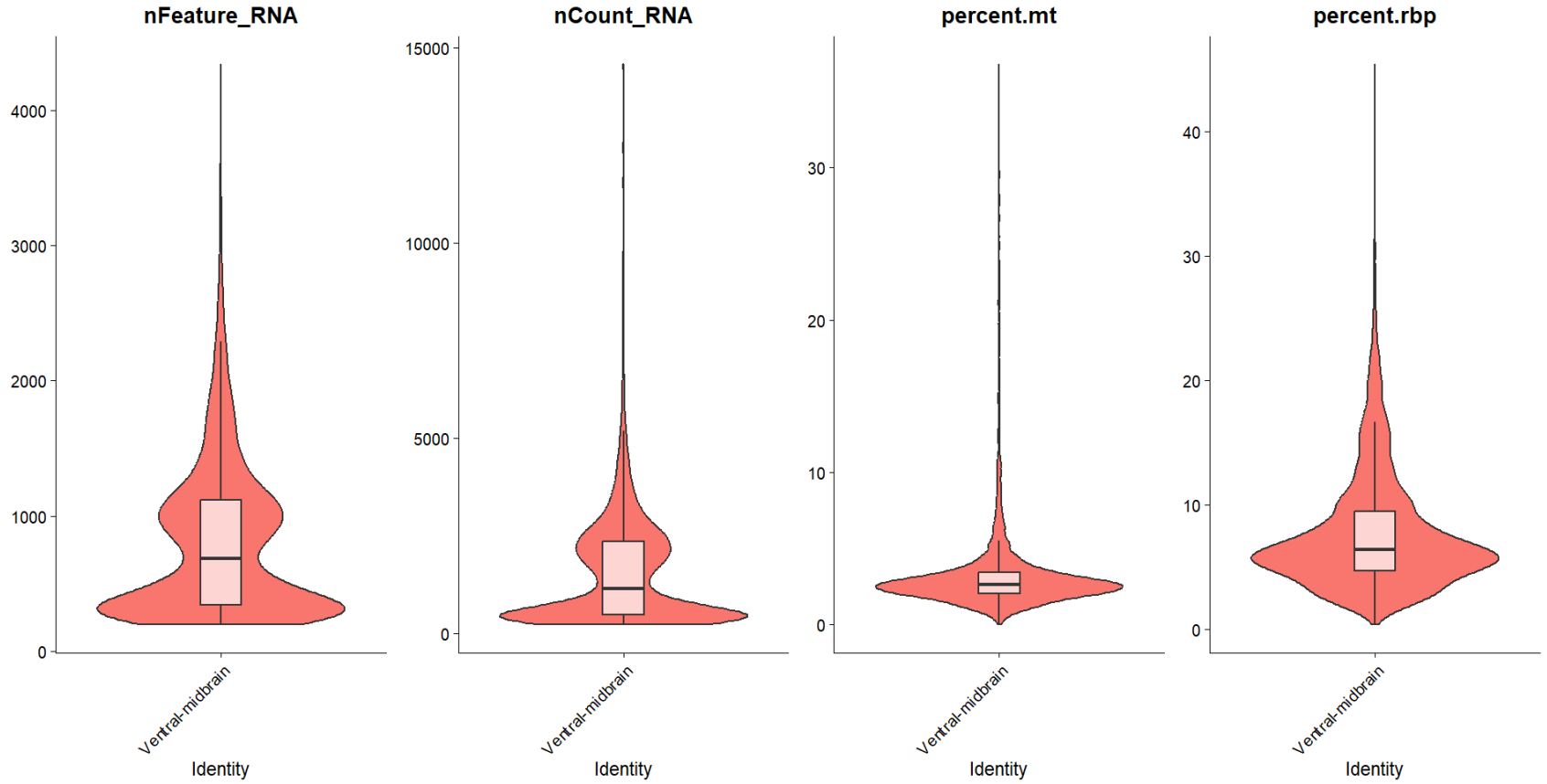
Cell type	Marker Genes
Dopaminergic neurons	Th, Slc6a3, Slc18a2
GABAergic neurons	Gad1, Gad2, Slc32a1
Astrocytes	Agt, Aqp4, Gja1
Oligodendrocytes	Opalin, Mbp, Mog, Cldn11
Microglia	Cx3cr1, Tmem119, C1qa
Endothelial cells	Cldn5, Bmx
COPs	Gpr17

5,812 cells

Filtering

3,292 cells

Quality control

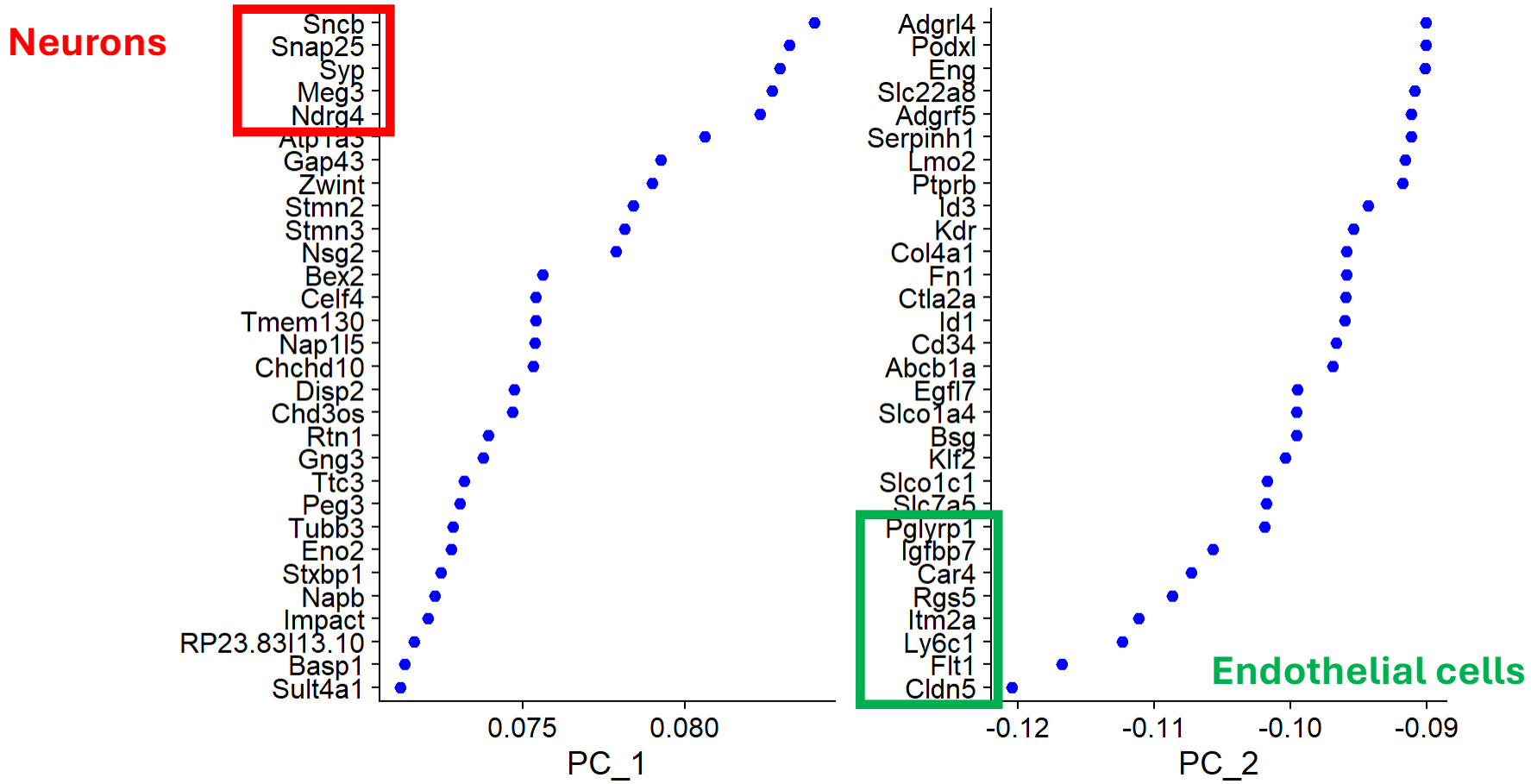


Seurat Object (5,812)
min.Cells = 3
Number of genes > 200

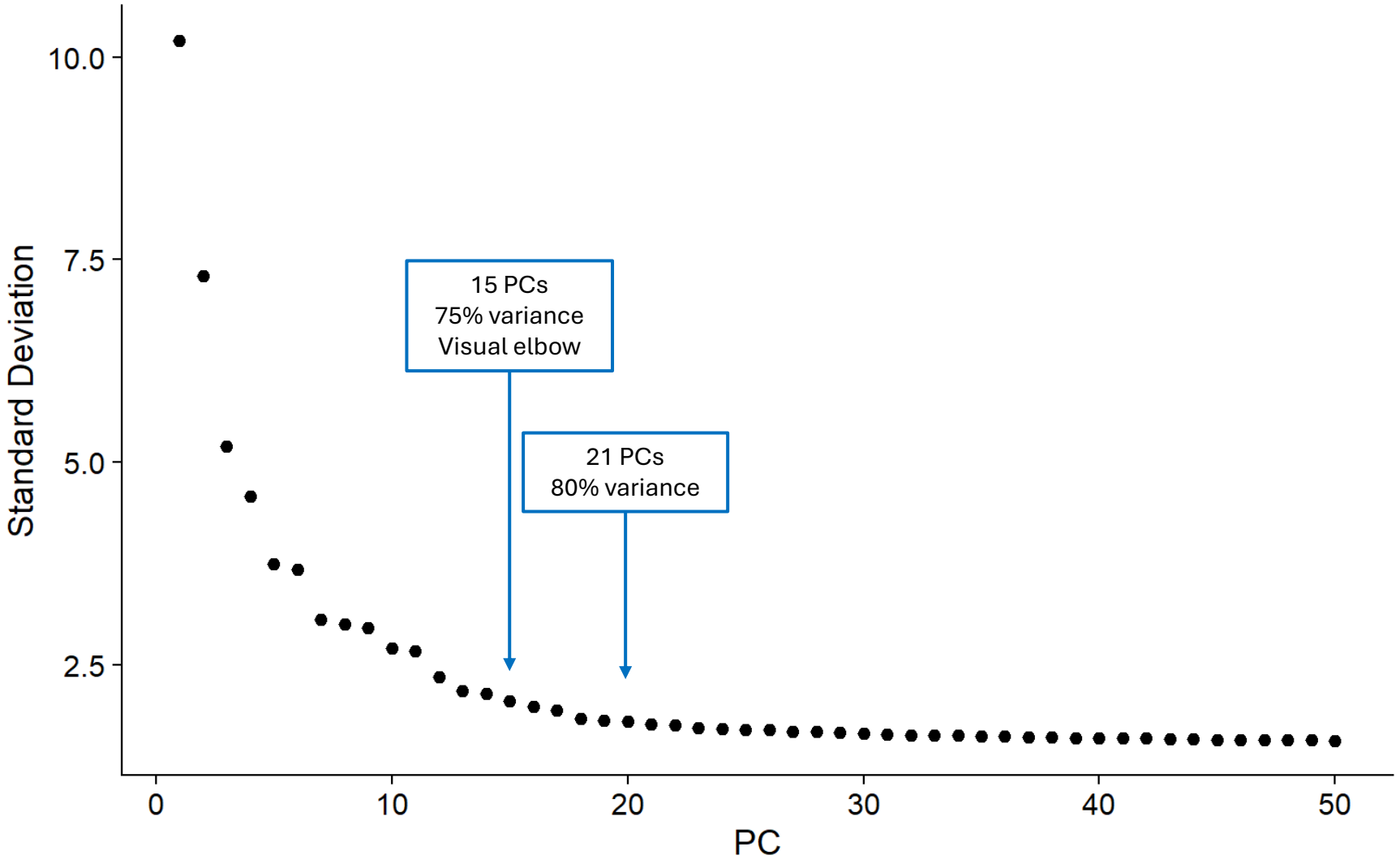
Quality filtering
500 < Number of genes < 3500
Mitochondrial RNA % < 5

Final number of cells
3,292 kept, ~56.6%
2,520 discarded, ~43.4%

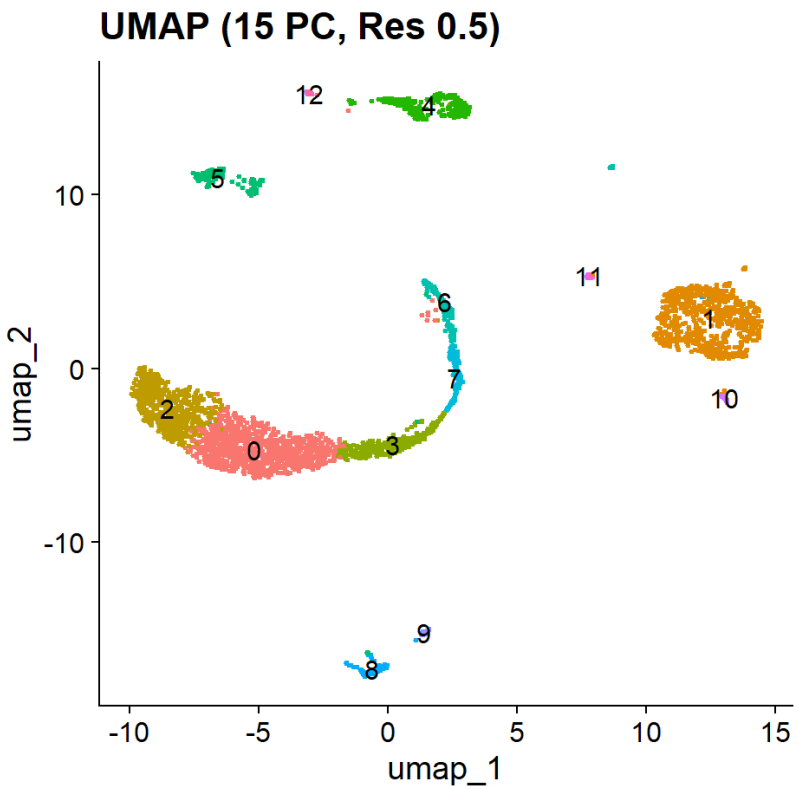
PCA – First two components

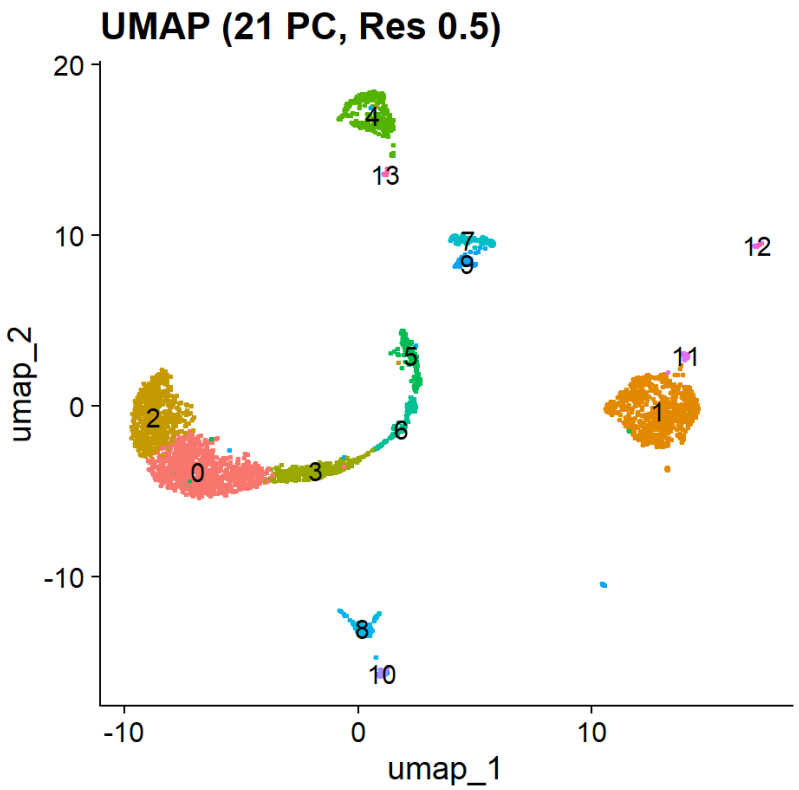


Elbow Plot


10.0
7.5
5.0
2.5
Standard Deviation
15 PCs
75% variance
Visual elbow
21 PCs
80% variance
0
10
20
30
40
50
PC

Clustering

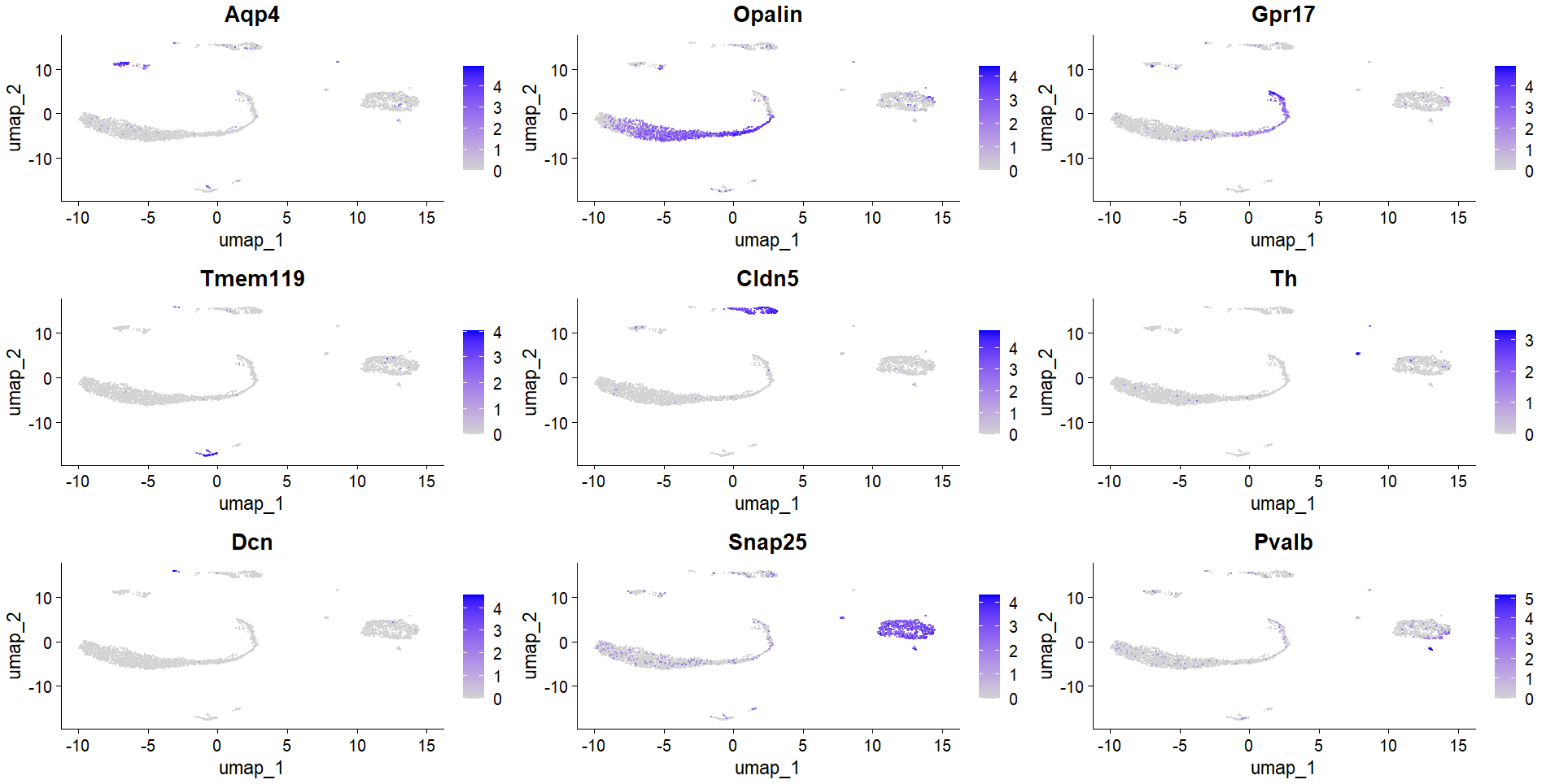




PC15	0	1	2	3	4	5	6	7	8	9	10	11	12	Tot
	896	692	564	262	247	167	159	114	87	34	28	23	19	3292

PC21	0	1	2	3	4	5	6	7	8	9	10	11	12	13	Tot
	882	693	571	257	246	154	115	106	87	78	34	27	23	19	3292

Possible Marker Genes Expression

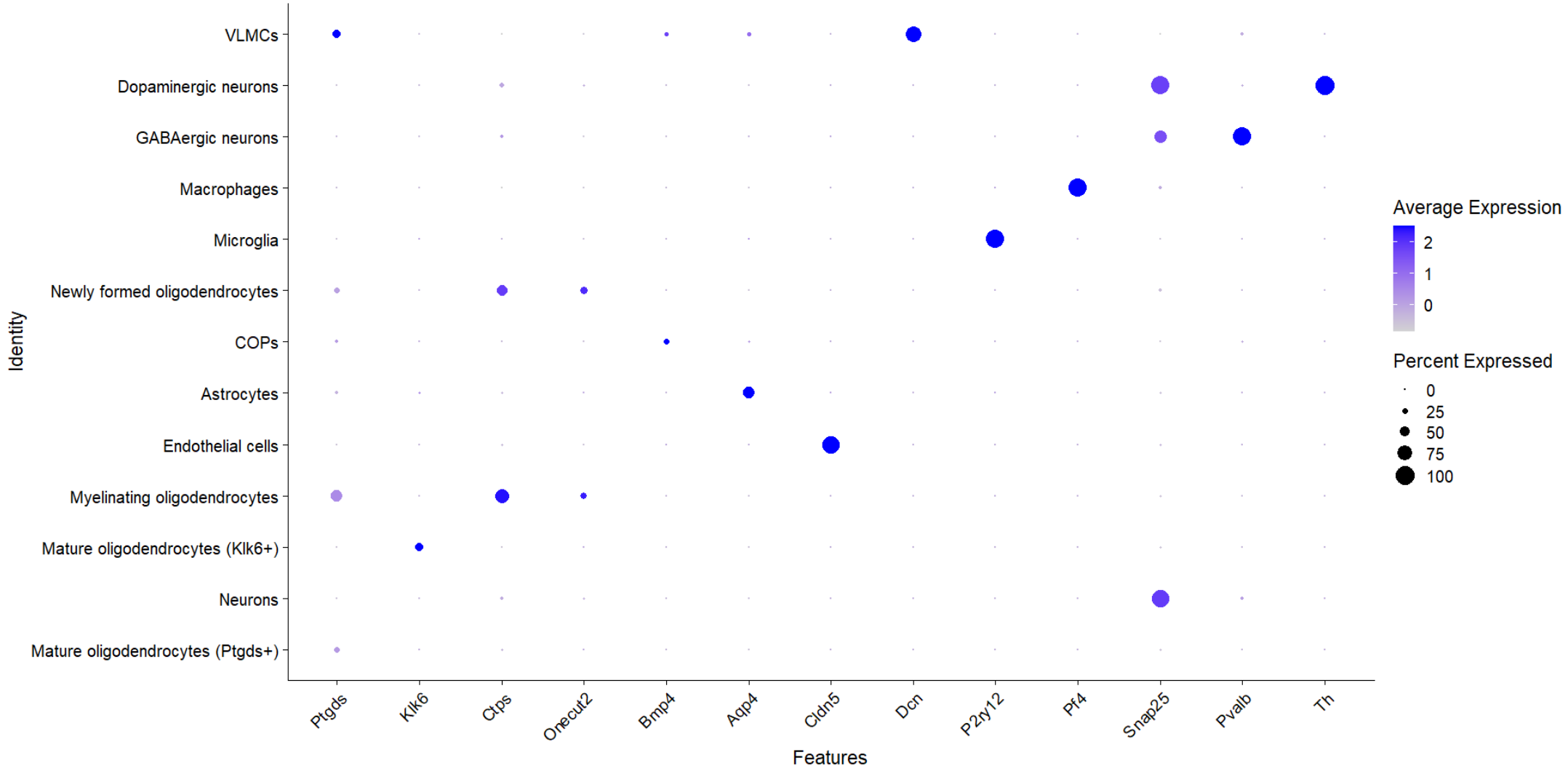


Cell type	Marker Gene
Astrocytes	Aqp4
Oligodendrocytes	Opalin
COPs	Gpr17
Microglia	Tmem119
Endothelial cells	Cldn5
Dopaminergic neurons	Th
VLMCs	Dcn
Neurons	Snap25
GABAergic neurons	Pvalb

Heatmap


0
1
2
3
4
5
6
7
8
9
10
11
12
Mal
Car2
Efhd1
Qdpr
Aspa
Htr2c
Vstm2l
Nrn1
Plvap
Vgf
Klk6
Pmp22
RP23.81P21.1
Plin3
Cdkn1c
Wfdc18
Ctps
Tmem141
RP23.61A11.2
Fam214a
Slc38a5
RP23.381H23.2
Pglyrp1
Abcb1a
Cldn5
Gfap
Agt
Acsbg1
Aqp4
Ntsr2
Neu4
Bmp4
Vcan
Mycl
Tmem108
Tmem2
RP23.47H16.3
Rras2
Cnksr3
Fam107b
P2ry12
Slc2a5
Tmem119
Selplg
Trem2
Ccl24
Cd209f
Ms4a7
Clec4n
Pf4
Il22
Sla
Pvalb
Tfap2b
Cnpy1
Chrna6
Slc6a3
Th
Ntf3
Trpc6
Itih2
Mfap4
Dcn
Fam180a
Lum

Marker Genes Expression Dotplot



Final result

